Chapter 1: Introduction and functions

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Voiceover: The next accessory organ we're going to talk about in the GI tract is the liver. The liver, and that's this big fatty guy right here.

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Now, the liver has to be one of the coolest organs in the body.

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It's the only one that you can take a piece of, this small chunk right here, and transplant into a recipient that has cirrhosis

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or liver failure, and it can grow to become a much larger organ that can sustain the patient. How wild is that?

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And so to better understand the liver, we'll start by discussing the main functions it's responsible for.

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The first thing we need to think about is that it's responsible for metabolism, metabolism, and that involves the breakdown of nutrients

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or catabolism, and I'll just write cat right here, catabolism, as well as the buildup of larger macromolecules, and that's through anabolism.

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So, catabolism and anabolism are achieved in the liver.

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The other function the liver is responsible for is storage of these nutrients.

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\When we're not breaking them down or building them up, we need to have a place for them to hang out, and so, they're responsible for the storage

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of our carbohydrates, so I'll just write carbs right here, and also mainly our fats, so these two guys, and they're stored in three different ways.

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First, carbs or fats can be stored as glycogen.

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Glycogen is just a polysaccharide, so a linkage of a whole bunch of monosaccharides or simple sugars.

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The other thing that we can store our carbs or our fats as are lipoproteins, lipo meaning fatty

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and proteins meaning, well, protein, and then lastly, we can also store these guys as triglycerides,

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triglycerides, which is one of the forms of fat that we ingest in our food.

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Now, these are two of the major macromolecules we use for energy.

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The other major macromolecule class that the liver sees are proteins, but these aren't really stored in the liver.

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Instead, they're processed into molecules like albumin and then cast off into the bloodstream, where they'll float around and carry out other functions,

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until they need to be retrieved by the liver to be broken down or used for other molecules.

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The other major process the liver is responsible for is called detoxification, detoxification,

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and this is one of those awesome scientific terms where the function is in the name.

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This is where we take toxins and modify them so they can't harm our body.

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This is achieved mainly by cytochrome P450 enzymes,

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cytochrome P450, and these are kind of funky enzymes, because unlike the other ones you've probably

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heard about before, these don't bind a specific single substrate or molecule.

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If you remember an enzyme like phenylalanine hydroxylase, that's something that will add a hydroxyl group to phenylalanine

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and only phenylalanine, but cytochrome P450's will take a whole bunch of different substrates and react with them.

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That's something that's pretty unique to this class of enzymes.

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Now, because of these cytochrome P450 enzymes, we have a problem when we take medications.

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In fact, we have a decrease in drug efficacy because of this detoxification process,

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and what we mean by drug efficacy is the amount of an effect a drug is able to have, and so, if the liver or these enzymes

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perceive medications that we ingest to be foreign or like toxins, they're going to try and break them down so they don't cause an effect to our body.

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So, doctors knows this, and because of that, a specific dose must be given to account for the amount of a drug that's going to be detoxified by the liver.

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And finally, as we'll talk about in a separate video in better detail, the last thing that the liver is responsible for is bile production,

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bile production, and as you might recall from our discussion on small bowel, bile is needed for the absorption of fats from our food.

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Okay, so now that we understand what are the main functions of the liver, how does it get all the nutrients that we ingest

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and absorb from our intestinal tract to be metabolized or stored or detoxified in the liver? Well, let's take a look at the blood supply to the liver.

Chapter 2: Liver blood supply and bile

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So, one of the things that makes the liver so unique is that it has two separate blood supplies.

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The first, that comes in from the intestinal tract, is from the portal vein.

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The portal venous system is another term for it, and the portal vein supplies the liver with nutrient-rich blood, nutrient-rich.

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So, where do these nutrients come from?

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Well, food that's absorbed in the intestinal tract will then go through the circulation and end up in the portal vein, to be delivered to the liver

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for metabolism, and so we consider this blood to be nutrient-rich, but where does the liver get its oxygen from?

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Well, that's a separate blood source, and that's actually through what's called the proper hepatic artery, the proper hepatic artery,

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and as the name suggests, this is arterial blood that will be supplying oxygen-rich red blood cells.

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So great, these are the two different sources of blood that go to the liver. Well, what about blood that leaves the liver?

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There's one main vessel that carries blood out of the liver, and that's called the hepatic vein.

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The hepatic vein is the exact opposite as the two other types of vessels we just talked about.

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It's nutrient- and oxygen-poor.

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Blood that leaves through the hepatic vein from the liver will circulate back to the heart to receive oxygen, flow past the intestines

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to receive nutrients, and then return to the liver, either through the portal vein or through the proper hepatic artery.

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The other output of the liver, that I guess I should mention right now, we'll talk about in more detail in a separate video, is bile, and bile mainly leaves the liver

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through what's called the common hepatic duct, the common hepatic duct, common hepatic duct that will take bile.